Euroleague_Bball

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```
# Load data
library(EuroleagueBasketball)
library(readr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   4.0.0     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.1.0

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
data(euroleague_basketball)

setwd("~/Desktop/TidyTuesday/Euroleague_Bball")
```

Which countries are most represented in the EuroLeague?

```
table(euroleague_basketball$Country) %>% sort()
```

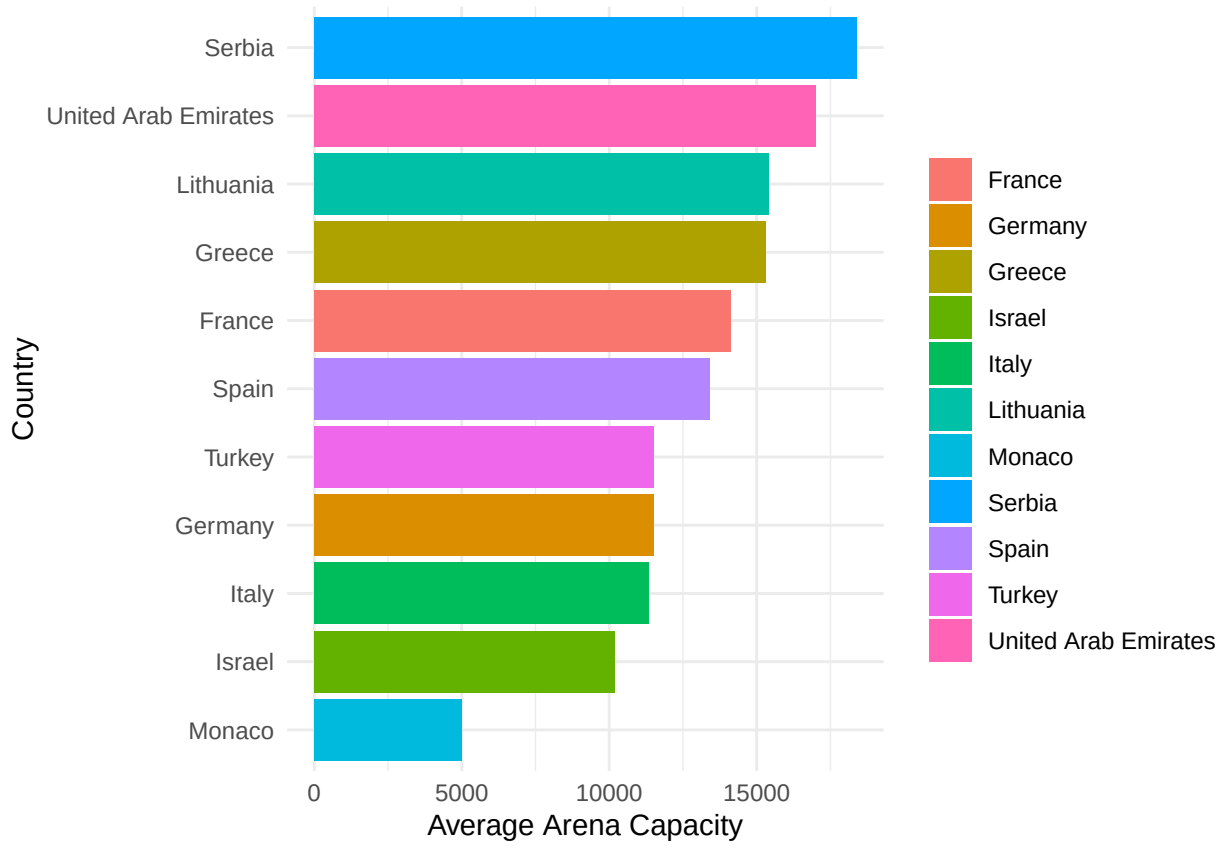
```
##
##              Germany            Lithuania               Monaco
##                    1                    1                    1
## United Arab Emirates               France               Greece
##                    1                    2                    2
##               Israel                Italy               Serbia
##                    2                    2                    2
##               Turkey                Spain
##                    2                    4
```

How do arena capacities compare across teams and countries? In R, the readr::parse_number() function might be helpful here.

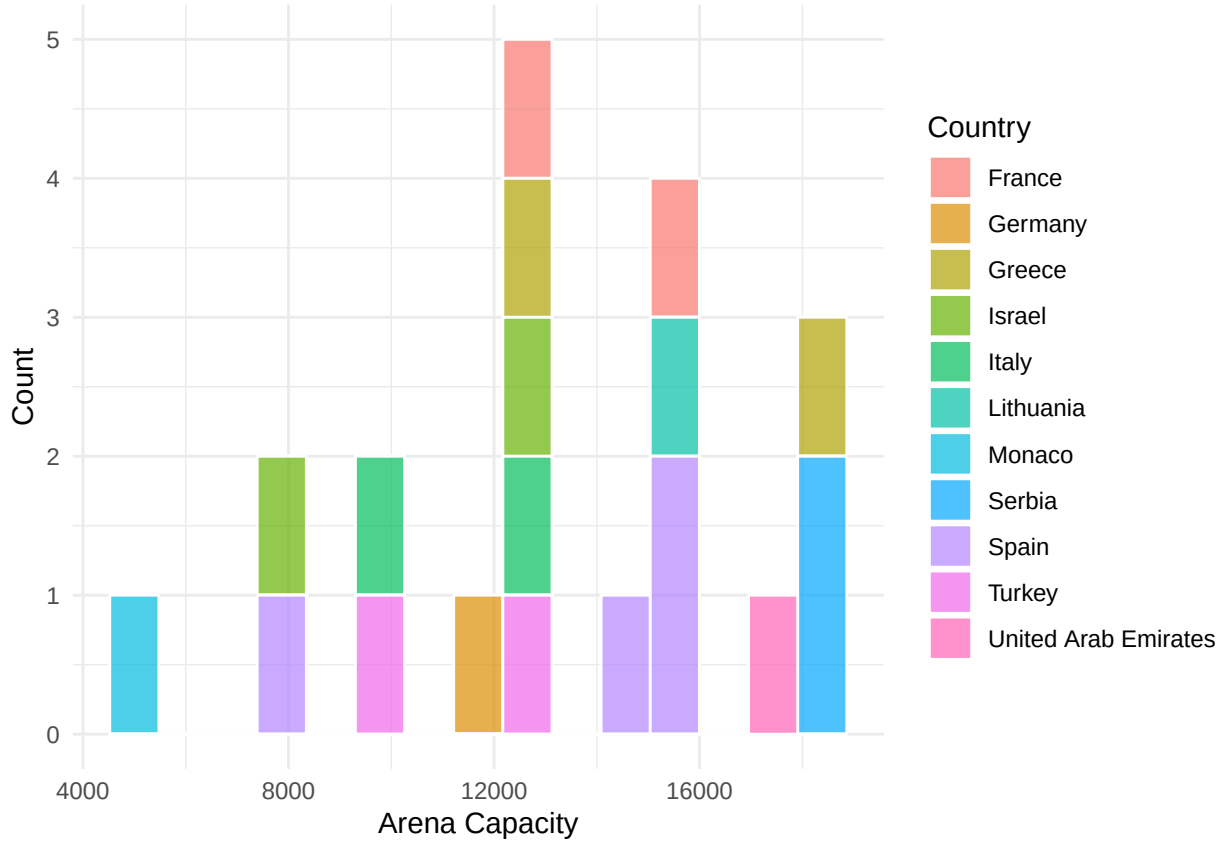
```
euroleague_basketball <- euroleague_basketball %>%
  mutate(Capacity = str_squish(Capacity)) %>%
  mutate(
    # extract all numeric values from the string
    Capacities = str_extract_all(Capacity, "\\d{1,3}(?:,\\d{3})*")
  ) %>%
  mutate(
    Capacity_old = parse_number(sapply(Capacities, `[`, 1)),
    Capacity_new = parse_number(sapply(Capacities, `[`, 2)),
    arena_capacity_high = pmax(Capacity_old, Capacity_new, na.rm = TRUE)
  ) %>%
  select(-Capacities)

#By country

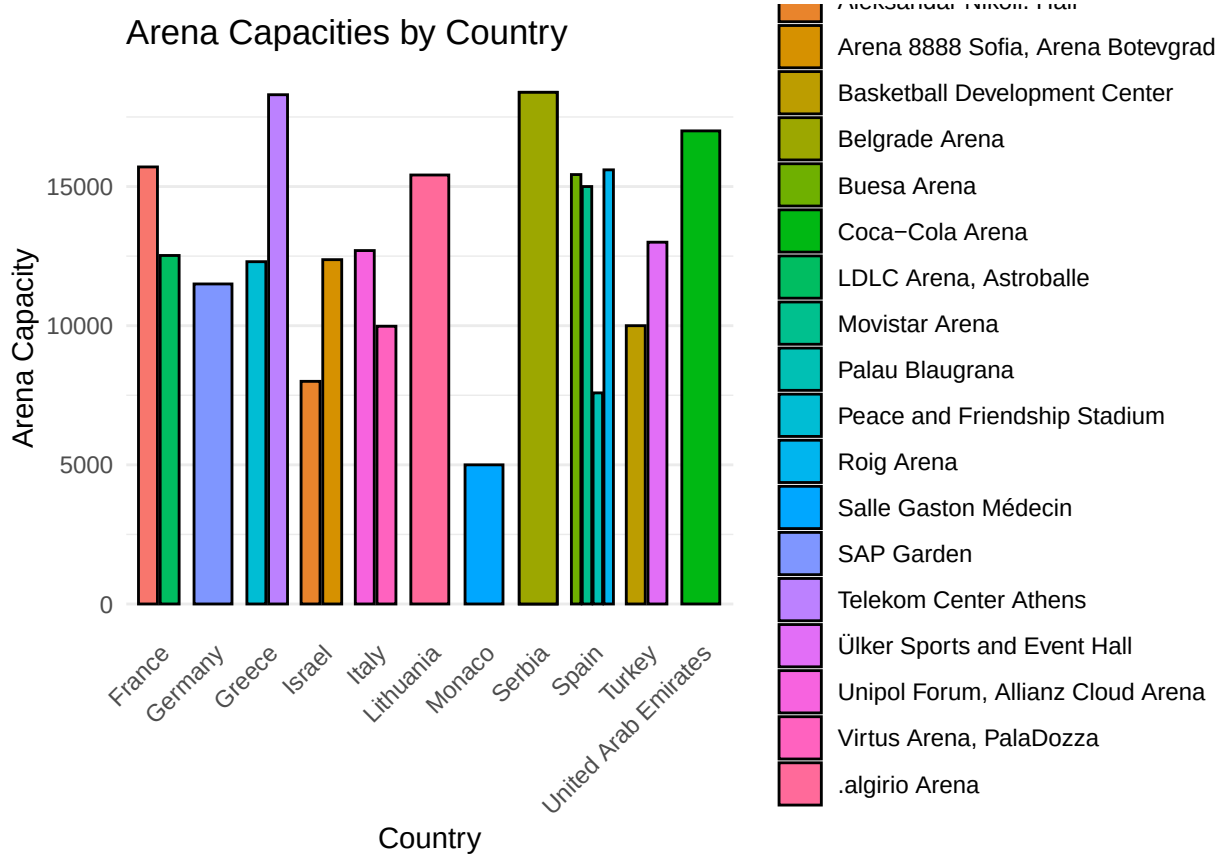
euroleague_basketball %>%
  group_by(Country) %>%
  summarise(mean_capacity = mean(arena_capacity_high, na.rm = TRUE)) %>%
  ggplot(aes(x = reorder(Country, mean_capacity), y = mean_capacity, fill = Country)) +
  geom_col() +
  coord_flip() +
  theme_minimal() +
  labs(x = "Country", y = "Average Arena Capacity", fill = "")
```




```
euroleague_basketball %>%
  ggplot(aes(x = arena_capacity_high, fill = Country)) +
  geom_histogram(bins = 15, alpha = 0.7, color = "white") +
  theme_minimal() +
  labs(x = "Arena Capacity", y = "Count")
```




```
euroleague_basketball %>%
  ggplot(aes(x = Country,
             y = arena_capacity_high,
             fill = Arena,     # each arena gets its own color
             group = Arena)) +
  geom_col(position = position_dodge(width = 0.8), width = 0.7, color = "black") +
  theme_minimal() +
  labs(
    x = "Country",
    y = "Arena Capacity",
    fill = "Arena",
    title = "Arena Capacities by Country"
  ) +
  theme(
    axis.text.x = element_text(angle = 45, hjust = 1),
    panel.grid.major.x = element_blank()
  )
```



Which clubs have been the most successful historically?

```
titles <- euroleague_basketball %>% select(Team, Titles_Won) %>% column_to_rownames(var = "Team") %>% a
final_four <- euroleague_basketball %>% select(Team, FinalFour_Appearances) %>% column_to_rownames(var =
final_four$FinalFour_Appearances <- as.numeric(final_four$FinalFour_Appearances)
final_four <- final_four %>% arrange(desc(FinalFour_Appearances))

print(titles)
```

```
##                             Titles_Won
## Panathinaikos                        7
## Real Madrid                          4
## Olympiacos                           3
## Anadolu Efes                         2
## Fenerbahçe                           2
## Virtus Bologna                       2
## Partizan                             1
## Žalgiris                             1
## Barcelona                            0
## Baskonia                             0
## Bayern Munich                        0
## Crvena zvezda Meridianbet            0
## Dubai Basketball                     0
## Hapoel IBI Tel Aviv                  0
## LDLC ASVEL                           0
## Maccabi Rapyd Tel Aviv               0
```

```
## Monaco                              0
## Olimpia Milano                      0
## Paris Basketball                    0
## Valencia Basket                     0
```

```r
print(final_four)
```

```
##                            FinalFour_Appearances
## Olympiacos                                    14
## Panathinaikos                                 13
## Real Madrid                                   12
## Fenerbahçe                                     7
## Anadolu Efes                                   5
## Virtus Bologna                                 5
## Olimpia Milano                                 4
## Partizan                                       3
## Monaco                                         2
## Žalgiris                                       2
## Barcelona                                      0
## Baskonia                                       0
## Bayern Munich                                  0
## Crvena zvezda Meridianbet                      0
## Dubai Basketball                               0
## Hapoel IBI Tel Aviv                            0
## LDLC ASVEL                                     0
## Maccabi Rapyd Tel Aviv                         0
## Paris Basketball                               0
## Valencia Basket                                0
```

```r
#If you needed to convert the years into individual numbers
euroleague_basketball <- euroleague_basketball %>%
  mutate(
    Titles_Won = case_when(
      is.na(Years_of_Titles_Won) | str_trim(Years_of_Titles_Won) == "" ~ 0L,
      str_detect(Years_of_Titles_Won, regex("none", ignore_case = TRUE)) ~ 0L,
      TRUE ~ str_count(Years_of_Titles_Won, "\\d{4}")
    )
  )

colnames(final_four)
```

```
## [1] "FinalFour_Appearances"
```